



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/13

Paper 1 Further Pure Mathematics 1

May/June 2026

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.



- 1 The cubic equation $x^3 + bx^2 + cx + d = 0$, where b, c and d are constants, has roots α, β and γ . It is given that

$$\begin{aligned}\alpha + \beta + \gamma &= 5, \\ \alpha^2 + \beta^2 + \gamma^2 &= -19, \\ \alpha^{-1} + \beta^{-1} + \gamma^{-1} &= 11.\end{aligned}$$

- (a)** Find the values of b , c and d .

[5]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.



(b) Find the value of $\alpha^3 + \beta^3 + \gamma^3$.

[2]





- 2** The matrix $\mathbf{M}$ is given by $\mathbf{M} = \begin{pmatrix} \frac{1}{2}\sqrt{2} & -\frac{1}{2}\sqrt{2} \\ \frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

- (a) The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations in the x - y plane.

Give full details of each transformation, and make clear the order in which they are applied. [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



[5]

This image shows a full page of a handwriting practice worksheet. It consists of multiple rows of horizontal dotted lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.



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7

[illegible]



4 (a) By considering $(2r+1)^4 - (2r-1)^4$, use the method of differences to prove that

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2. \quad [6]$$

[illegible]



Let $S_n = 1^3 - 2^3 + 3^3 - 4^3 + \dots + (-1)^{n+1}n^3$.

- (b)** Show that $S_{2n} = n^2(an + b)$, where a and b are integers to be determined. [3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



5 The curve C has polar equation $r^2 = \theta \sec^2 \theta$, where $0 \leq \theta \leq \frac{1}{3}\pi$.

(a) Sketch C and state the greatest distance of a point on C from the pole.

[2]

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(b) Find the exact value of the area of the region bounded by C and the half-line $\theta = \frac{1}{3}\pi$.

[5]

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- (c) Show that C has Cartesian equation $y = x \tan(x^2)$, where $0 \leq x \leq \frac{1}{3}\sqrt{3\pi}$. [3]

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- (d) Using your answer to **5(b)**, deduce the exact value of $\int_0^{\frac{1}{3}\sqrt{3\pi}} x \tan(x^2) dx$. [3]

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6 The curve C has equation $y = \frac{x+1}{x^2+x+1}$.

(a) Show that C has no vertical asymptotes and state the equation of the horizontal asymptote. [2]

[illegible]

(b) Find the coordinates of any stationary points on C . [3]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(c) Sketch C , stating the coordinates of the intersections with the axes.

[3]

(d) Sketch the curve with equation $y = \left| \frac{x+1}{x^2+x+1} \right|$.

[2]



(e) Find in exact form the set of values of x for which $\left| \frac{x+1}{x^2+x+1} \right| < \frac{1}{3}$. [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

7 The plane Π has equation $\mathbf{r} = \mathbf{j} + 2\mathbf{k} + s(\mathbf{i} - \mathbf{j}) + t(-3\mathbf{i} + \mathbf{j} - \mathbf{k})$.

(a) Obtain an equation of Π in the form $ax + by + cz = d$.

[4]

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(b) The point P has position vector $(6 + 6\theta)\mathbf{i} + (1 + \theta)\mathbf{j} + (2 + 5\theta)\mathbf{k}$.

Find the set of values of θ for which the perpendicular distance of P from Π is less than $\sqrt{6}$. [4]

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- (c) Find an equation for the common perpendicular to the lines $\mathbf{r} = \lambda(\mathbf{j} + 2\mathbf{k})$ and $\mathbf{r} = \mathbf{i} + 2\mathbf{k} + \mu(-3\mathbf{i} + \mathbf{j} - \mathbf{k})$.

[8]

[illegible]



If you use the following page to complete the answer to any question, the question number must be clearly shown.

[illegible]







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